

KINGSVILLE NAS, TX, USA

WMO: 722516

Lat: 27.500N

Lon: 97.817W

Elev: 50

StdP: 14.67

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 33.9	(c) 37.3	(d) 17.0	(e) 13.0	(f) 47.8	(g) 22.4	(h) 16.8	(i) 50.6	(j) 28.0	(k) 73.9	(l) 25.3	(m) 68.0	(n) 8.3	(o) 0	(p) 0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 20.1	(c) 99.1	(d) 78.5	(e) 97.4	(f) 78.5	(g) 95.9	(h) 78.5	(i) 82.4	(j) 91.9	(k) 81.5	(l) 90.3	(m) 80.8	(n) 89.3	(o) 13.1	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.5	158.4	84.8	79.5	153.4	84.1	78.9	150.5	83.9	46.2	92.6	45.2	90.2	44.4	89.1	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.8	(b) 22.1	(c) 19.9	(e) 28.0	(f) 103.1	(g) 4.0	(h) 3.4	(i) 25.2	(j) 105.5	(k) 22.8	(l) 107.5	(m) 20.6	(n) 109.4	(o) 17.7	(p) 111.9		
(a) 24.8	(b) 22.1	(c) 19.9	(e) 25.3	(f) 83.9	(g) 3.6	(h) 1.5	(i) 22.8	(j) 85.0	(k) 20.7	(l) 85.9	(m) 18.6	(n) 86.8	(o) 16.0	(p) 87.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.3	57.6	62.1	67.0	73.8	80.0	84.3	85.6	86.0	81.8	74.9	66.0	59.8
	DBStd	12.22	9.58	9.82	8.73	6.66	4.79	3.11	2.52	2.72	4.08	6.78	8.92	9.64
	HDD50	88	38	17	6	0	0	0	0	0	0	0	5	22
	HDD65	845	267	156	84	12	0	0	0	0	0	10	95	221
	CDD50	8598	274	354	534	715	929	1030	1105	1116	955	773	486	327
	CDD65	3881	38	74	147	277	464	580	640	651	505	319	125	61
	CDH74	41188	261	498	913	2229	4673	6930	7978	8014	5288	3028	995	381
	CDH80	18819	47	138	239	774	1894	3440	4153	4207	2420	1164	270	73
(14) Wind	WSAvg	9.7	9.4	10.0	10.7	11.3	11.3	10.3	10.4	9.1	7.3	8.3	8.9	8.9
(15) Precipitation	PrecAvg	28.0	1.4	1.6	0.9	1.6	2.9	3.3	2.3	3.0	4.3	3.2	1.3	1.4
	PrecMax	49.7	3.8	4.1	4.6	8.2	11.3	12.4	12.0	20.9	19.4	13.7	5.0	8.7
	PrecMin	12.7	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.3	0.1	0.0	0.0
	PrecStd	10.2	0.9	1.3	1.0	1.8	2.5	2.7	2.6	3.7	3.9	3.3	1.2	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	89.8	90.4	96.5	96.7	100.0	100.4	101.5	98.9	93.7	90.1	85.5
		MCWB	67.7	68.2	71.2	74.1	77.1	78.5	78.6	78.1	76.6	76.8	74.4	70.7
	2%	DB	80.6	83.9	85.5	90.8	93.4	96.9	98.3	99.3	96.1	91.1	86.2	82.0
		MCWB	67.0	69.4	71.1	72.7	77.2	78.9	78.7	79.0	77.9	76.6	72.8	69.6
	5%	DB	76.7	80.3	82.2	87.4	91.2	95.1	96.8	97.6	93.4	89.1	83.0	78.7
		MCWB	65.5	68.0	70.3	73.1	76.8	78.6	78.8	78.9	77.2	75.8	71.7	68.5
	10%	DB	72.8	76.1	79.3	84.3	89.2	93.2	94.8	95.7	91.1	86.8	79.9	74.7
		MCWB	64.3	66.4	68.5	72.3	76.4	78.1	78.9	78.5	77.3	74.5	70.3	68.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.5	73.7	76.1	80.2	82.5	83.7	83.5	83.7	82.2	81.1	78.1	74.7
		MCDB	78.0	81.8	84.1	86.7	89.6	93.1	94.7	94.8	90.0	87.6	84.2	80.6
	2%	WB	70.6	72.1	74.2	77.7	80.5	82.1	82.2	82.0	81.0	79.7	76.3	72.8
		MCDB	75.6	79.4	81.4	85.3	87.4	91.4	91.8	91.9	88.7	85.8	81.6	77.8
	5%	WB	69.0	70.7	72.8	76.1	79.4	81.1	81.2	81.0	80.2	78.4	74.6	71.3
		MCDB	74.1	76.6	78.8	83.2	86.5	90.0	90.0	90.2	87.6	84.6	79.8	75.7
	10%	WB	66.9	69.2	71.5	74.8	78.2	80.2	80.3	80.2	79.3	77.4	72.6	69.6
		MCDB	71.8	73.7	75.9	81.1	85.3	88.3	89.0	89.2	86.6	83.7	77.8	73.9
(35) Mean Daily Temperature Range	5% DB	MDBR	21.7	21.3	20.1	20.1	18.1	18.7	19.5	20.1	19.1	21.3	21.4	20.5
		MCDBR	25.6	25.2	21.7	22.1	20.0	20.6	22.0	22.6	21.8	21.2	22.7	23.5
		MCWBR	14.9	12.8	10.4	9.6	6.9	5.9	5.9	5.8	6.7	9.0	11.3	13.4
	5% WB	MCDBR	20.8	20.3	18.0	17.9	16.0	17.5	19.1	20.2	18.1	17.7	19.7	19.5
		MCWBR	13.5	11.2	9.6	8.3	6.8	6.5	6.2	6.2	6.4	8.6	11.3	12.8
(40) Clear-Sky Solar Irradiance	taub		0.358	0.368	0.385	0.431	0.470	0.447	0.453	0.449	0.439	0.387	0.374	0.351
	taud		2.455	2.429	2.384	2.267	2.190	2.314	2.316	2.324	2.320	2.456	2.442	2.489
	Ebn at Noon		281	287	288	276	264	268	266	267	267	277	273	278
	Edh at Noon		31	34	38	44	47	42	41	41	40	33	31	29
(44) All-Sky Solar Radiation	RadAvg		972	1158	1454	1730	1902	2113	2097	2010	1656	1441	1086	871
	RadStd		105	195	170	168	151	99	165	110	143	157	114	124

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.79	N/A	N/A	+1.03	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	+0.63	N/A	N/A	N/A	N/A	+218	+202

Nomenclature: See separate page